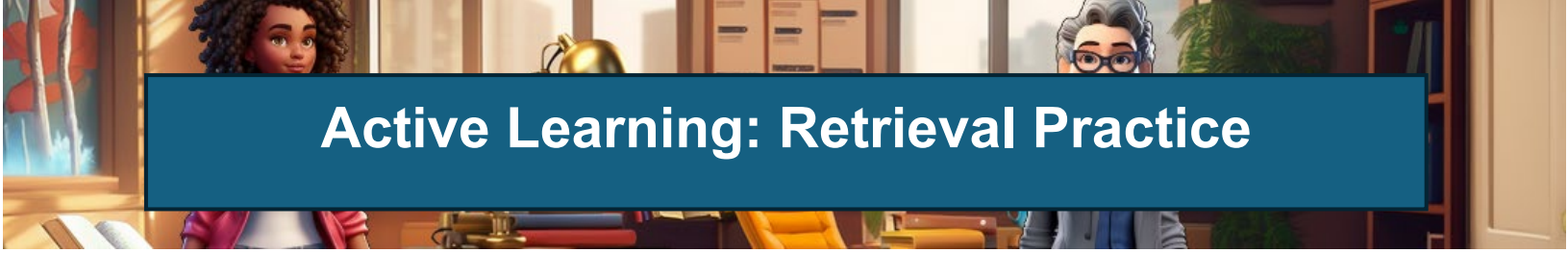

Active Learning: Retrieval Practice

Active learning improves memory strength and understanding. It is “active” because it involves the learner in the process through questioning, applying concepts, etc. It helps more than passive learning which usually involves listening or reading without interaction (e.g., rereading notes). Here are some ways to engage in retrieval using active learning techniques.

Active recall.

- ⌘ Retrieve information from memory without looking at notes or books.
- ⌘ It is more powerful than just reviewing notes.

Students who used this strategy scored over 50% higher on tests than students who just reviewed their notes (Dunlosky et al. (2013))

Step 1: *Study a topic or read a chapter thoroughly.*

Step 2: *Close your book or cover your notes.*

Step 3: *Write or speak everything you remember about the topic.*

Step 4: *Check your answers and identify what you missed.*

Step 5: *Review weak areas and repeat the recall exercise later.*

Feynman Technique.

- ⌘ The learner explains concepts in their own words as if teaching a beginner.
- ⌘ It forces simplification and clarity, which exposes gaps in comprehension and facilitates deeper learning.

Step 1: *Choose a concept or topic you want to learn.*

Step 2: *Write an explanation of the concept as if you’re teaching it to someone with no prior knowledge.*

Step 3: *Identify areas where your explanation is lacking or unclear.*

Step 4: *Go back to the source material to review those gaps.*

Step 5: *Refine and simplify your explanation until it’s crystal clear.*

Sources: Academic Square: https://academiasquare.com/studying/study-techniques/

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Active Learning: Retrieval Practice

Method of Loci.

- ⌘ Also called the memory palace technique
- ⌘ It enhances recall by associating pieces of information with physical locations you mentally navigate. .

Useful for memorizing lists or sequences

Step 1: *Visualize a familiar location (e.g., your house).*

Step 2: *Identify specific, sequential "stops" or landmarks in this location.*

Step 3: *Associate each piece of information with a specific stop.*

Step 4: *During recall, mentally walk through the space and retrieve the information placed at each stop.*

Step 5: *Practice the walk until it becomes fluent and the associations stick.*

Rote learning.

- ⌘ Involves continuous repetition.
- ⌘ Can be useful for recalling foundational facts such as vocabulary, formulas, or dates.

It is most effective when combined with mnemonic techniques.

Step 1: *Select factual content you need to memorize (e.g., terms, formulas).*

Step 2: *Repeat the content out loud or in writing several times.*

Step 3: *Use flashcards or apps for regular drills.*

Step 4: *Schedule reviews using spaced repetition intervals.*

Step 5: *Test yourself frequently to reinforce memorization.*

Cognitive learning theory

- ⌘ Encourages learners to actively interpret and organize new information.
- ⌘ useful for meaningful learning rather than rote memorization.

Step 1: *Connect new concepts to prior knowledge or real-life experiences.*

Step 2: *Organize material using mental maps or diagrams.*

Step 3: *Reflect on what strategies helped or hindered learning.*

Step 4: *Adapt study approaches based on feedback and outcomes.*

Step 5: *Engage in discussions or teaching to reinforce understanding.*

Sources: James Madison University

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https://sites.google.com/view/gears-game/